

# Reading list for MATLAB course

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These books are all simply suggestions, none of them are required for the course. It is the case with most of these books that they will not help you directly with this course. All the books listed here are for further reading about subjects talked about in the course.

## Computer Science books

Introduction to the theory of computation - Michael Sipser available here This book gives one an over view of the all the theory necessary for a BS in computer science. This is my favorite book on CS theory as it speaks to all the critical subjects in a very readable way.

Elements of Programming - Stepanov available here This is a difficult read, that said it is the most influential programming book I have ever read. This is not worth reading unless you have some knowledge of computer science theory.

An introduction to parallel programming - Pacheco available here This is overkill for this class but it explain in more depth some of the automated calculus that MATLAB is doing.

## Math books

No Bullshit guide to linear algebra - Savov available here This is an excellent reference guide to linear algebra. It talks through all the necessary math leading to linear algebra and even expands past it a bit. This is written

from the perspective of a computer scientist, as such most of the problems are targets at being solved in discrete space.

Ordinary differential equations - Tenenbaum available [here](#) This book is the crowd favorite for reference material on ODEs.

Calculus - Spivak available [here](#) If I had to choose one reference book for calculus it would be this one. The detail is amazing and its a crowd favorite among mathematicians. It is difficult to read and quite expensive though. In our case this is for integral calculus examples.